

MAXWELL BURSEY

St. John's, NL · (709) 636-8902 · maxwellb@mun.ca · linkedin.com/in/maxwellbursley · maxwellbursley.github.io

EDUCATION

B.Eng. in Electrical Engineering (Co-op) — Term 5 of 8

2023 – 2028

Memorial University of Newfoundland — St. John's, NL

GPA: 3.87 / 4.0 (85.5%) Expected Graduation: May 2028 Honours Graduate, Corner Brook Regional High School (95%)

EXPERIENCE

Co-op Engineering Student

Aug. – Dec. 2025

The Commons, Queen Elizabeth II Library, MUN — St. John's, NL

- Independently owned a full-cycle workstation usage-tracking project: prototyped PIR motion-detection and AC current-sensing circuits, developed noise-reduction classification logic, and delivered formal findings to supervisors.
- Produced a comprehensive technical report and presentation addressing scalability, data privacy, and operational constraints; maintained weekly reporting cadence aligned to institutional requirements.
- Coordinated cross-functional workflows via Slack and Trello, ensuring timely issue escalation and on-schedule delivery across the full work term.

Head Server

Jul. 2022 – Aug. 2023

WOWOs Asian Homestyle Cooking — Corner Brook, NL

- Managed dining floor operations for a high-volume restaurant, maintaining consistently high guest satisfaction across a 13-month tenure.
- Trained and onboarded new service staff; configured POS payment systems and digital menus, applying technical skills in a live production environment.

Equipment Operator & Office Assistant

Jun. – Sep. 2021

Trademark Lumber Company — Deer Lake, NL

- Operated industrial lumber processing equipment under strict safety protocols, contributing to a zero-incident season across the full work term.
- Processed employee timecards and calculated weekly payroll with 100% accuracy, ensuring timely compensation for all staff; supported yard logistics and inventory tracking.

PROJECTS

Analog Voltage Reference PCB Design

Jan. – Feb. 2026

Personal / Academic Engineering Project

- Designed a two-layer PCB delivering a stable 5 V analog reference from a 3-cell LiPo battery, implementing reverse polarity protection, NCPI117 linear regulation, and an MCP6001 op-amp voltage follower for low-impedance, low-noise output.
- Created full schematic and PCB layout in KiCad with custom battery connector footprints, dedicated power nets, and a continuous ground plane; completed ERC/DRC verification, Gerber/drill file generation, detailed BOM, and 3D visualization for manufacturing readiness.

Laptop Station Usage Tracking System

Oct. – Dec. 2025

Work Term Project — The Commons, MUN

- Designed and benchmarked an automated sensing system classifying active vs. inactive shared workstations; produced a formal technical report covering privacy, scalability, and maintenance recommendations for supervisor decision-making.

Automatic Irrigation System

Feb. – Apr. 2024

Personal Project — St. John's, NL

- Built a closed-loop microcontroller-based irrigation system integrating capacitive soil moisture sensors, a relay-driven pump, and LCD status display; programmed threshold control logic in C and calibrated sensors for consistent field performance.

SKILLS & CERTIFICATIONS

Programming

Python | C | C++

Engineering Software

KiCad | MATLAB | SolidWorks | Onshape | PSpice

Productivity & Tools

Microsoft Office 365 (Word, Excel, PowerPoint) | Slack | Trello

Core Competencies

PCB design & layout | Embedded systems | Sensor integration | Technical documentation

Certifications

WHMIS | Tech Stewardship Micro-Credential | Excel Essential Training | Full Class 5 Licence